[S06] Diffusion Models: From Adding Noise to Generating Images

TOPIC AREA Generative AI

I. Learning Objectives and Guiding Question

Explain forward noising and reverse denoising, including why latent diffusion reduces computational demands.

Guiding question: How can learning to remove noise enable a model to generate new images?

II. Required Tasks

1. Distinguish generative and discriminative models; qualitatively compare GANs, autoregressive models, and diffusion models.
2. Explain the forward process, β schedule, Gaussian noise, reverse process, noise predictor, and sampling. Draw a diagram illustrating both directions.
3. Manually calculate one step of one-dimensional forward noising using chosen values for x_0, β, and ε; define the symbols and interpret the stochastic expectation.
4. Illustrate noise addition to a small image (e.g., 28 × 28) at five noise levels. If resources permit, demonstrate denoising or image generation with a small pretrained model; training is not required.
5. Explain latent diffusion using a VAE encoder/decoder, operations in latent space, and cross-attention conditioning; illustrate the pipeline.
6. Discuss sampling speed, quality, prompt-based control, and limitations related to training data, copyright, and bias. Compare one latent-diffusion setting with pixel-space diffusion using explicit criteria.

III. Equations / Representations to Explain

$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{(1-\beta_t)}x_{t-1}, \beta_t I)$$

$$L_simple = E_{t,\,x0,\,\varepsilon}[\|\varepsilon - \varepsilon\theta(x_t, t)\|^2]$$

Student need to define every symbol, specify appropriate units/value ranges and modeling assumptions, and provide a complete step-by-step worked example.

IV. Minimum Expected Outputs

Image sequence with increasing noise, forward/reverse-process diagram, pixel-versus-latent comparison, and an image-generation example if available.

V. Required and Recommended Research Papers

Required: Ho, J., Jain, A. & Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. NeurIPS 2020.

https://papers.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html

Reading focus: Read the forward-noising/reverse-denoising processes and noise-prediction objective.

Recommended: Rombach, R. et al. (2022). High-Resolution Image Synthesis With Latent Diffusion Models. CVPR 2022, 10684–10695.

https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html

Reading focus: Read how diffusion in latent space differs from pixel-space diffusion.

VI. Format and Submission Requirements

- Team size: three students are recommended. Suggested preparation time: approximately two weeks after the topic is assigned.
- Submit a 6–10-page research-style report in Vietnamese on A4 paper (excluding references and appendices), organized as: Abstract, Introduction, Background/Paper Review, Method/Worked Example, Results and Discussion, Conclusion, and References.
- Prepare 8–12 content slides for a 15-minute seminar presentation followed by approximately 5 minutes of Q&A. Every team member must present and answer at least one question.
- Read all papers marked “Required” in this assignment. Explain the material in your own words and distinguish findings reported in the papers from examples or experiments developed by your team.
- Programming-based experiments are encouraged but not generally mandatory. A worked numerical example, an illustrative model, and an evidence-supported comparison are mandatory.

- Cite all figures, tables, datasets, equations, and AI tools used. Independently verify all claims suggested by AI-generated content.

VII. Seminar Assessment Rubric (100 points)

Assessment criterion	Requirements for full marks	Points
Problem statement and objectives	Establish relevant context, a clear research question, and an appropriate scope.	5
Reading and synthesis of research papers	Accurately explain the original papers' objectives, contributions, methods, and findings, with appropriate source comparison.	15
Explanation of algorithms and equations	Define all variables and assumptions, describe algorithmic steps, and provide at least one correct numerical example or diagram.	15
Evidence, comparison, or experimentation	Include an appropriate baseline or reference, verifiable data or examples, and clearly explained figures and tables.	10
Critical analysis and limitations	Discuss strengths, weaknesses, and conditions of applicability. Avoid overstating conclusions based on a small demonstration.	10
Slide quality and citations	Prepare 8–12 clear, legible slides, cite sources for figures/equations and relevant papers, and avoid verbatim copying.	10
Seminar presentation	Deliver a coherent presentation within 15 minutes, explain examples clearly, and ensure balanced participation.	15
Q&A and critical discussion	Answer questions accurately, justify technical decisions, and ensure that each member answers at least one question.	20

Performance levels for each criterion: Excellent (100%), Good (75%), Satisfactory (50%), and Unsatisfactory (0–25%) of the available points, depending on the quality of evidence submitted.